

Tutorial sheet Mid-2

ECE/CSE

Long Answer Questions:

1. Solve $x(y-z)p + y(z-x)q = z(x-y)$

Sol: Given PDE is $x(y-z)p + y(z-x)q = z(x-y)$

$$p = x(y-z) ; q = y(z-x) ; r = z(x-y)$$

The A.E.s are $\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r}$

$$\Rightarrow \frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$$

choose Multiplier (l, m, n) as $(1, 1, 1)$

$$\begin{aligned}\therefore \text{Each ratio} &= \frac{l dx + m dy + n dz}{lp + mq + nr} \\ &= \frac{1 \cdot dx + 1 \cdot dy + 1 \cdot dz}{x(y-z) + y(z-x) + z(x-y)} \\ &= \frac{dx + dy + dz}{xy - xz + yz - xy + xz - yz}\end{aligned}$$

$$\Rightarrow dx + dy + dz = 0$$

IOBS

$$\Rightarrow \int dx + \int dy + \int dz = \int 0$$

$$\Rightarrow \boxed{x + y + z = c_1} \text{ is one sol}$$

choose multiplier (l, m, n) as $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$

$$\begin{aligned}\therefore \text{Each ratio} &= \frac{l dx + m dy + n dz}{lp + mq + nr} \\ &= \frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{\frac{1}{x} \cdot x(y-z) + \frac{1}{y} \cdot y(z-x) + \frac{1}{z} \cdot z(x-y)} \\ &= \frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{0}\end{aligned}$$

$$\Rightarrow \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0$$

IOBS

$$\Rightarrow \int \frac{1}{x} dx + \int \frac{1}{y} dy + \int \frac{1}{z} dz = \int 0$$

$$\Rightarrow \log x + \log y + \log z = \log C_2$$

$$\Rightarrow \log(xyz) = \log C_2 \Rightarrow \boxed{xyz = C_2} \text{ is sol}$$

$$\therefore \text{The G. Sol is } \phi(C_1, C_2) = 0$$

$$\Rightarrow \phi(x+y+z, xyz) = 0.$$

2. Solve $(mz - ny)P + (nx - lz)Q = ly - mx$

Sol: Given PDE is $(mz - ny)P + (nx - lz)Q = ly - mx$

$$P = mz - ny; Q = nx - lz; R = ly - mx$$

$$\text{The A. Eqs are } \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}$$

Take (l, m, n) as (l, m, n) then

$$\text{Each Ratio} = \frac{l dx + m dy + n dz}{lP + mQ + nR}$$

~~$$\frac{l dx + m dy + n dz}{l(mz - ny) + m(nx - lz) + n(ly - mx)}$$~~

$$= \frac{l dx + m dy + n dz}{l(mz - ny) + m(nx - lz) + n(ly - mx)}$$

$$= \frac{l dx + m dy + n dz}{lmz - lny + mnx - lmz + lny - mnx}$$

$$= \frac{l dx + m dy + n dz}{0}$$

$$\Rightarrow l dx + m dy + n dz = 0$$

Integrating

$$\Rightarrow \int l dx + \int m dy + \int n dz = \int 0$$

$$\Rightarrow \boxed{lx + my + nz = C_1} \text{ is one sol.}$$

Take (l, m, n) as (x, y, z) then

$$\begin{aligned} \text{Each ratio} &= \frac{l dx + m dy + n dz}{l p + m q + n r} \\ &= \frac{x dx + y dy + z dz}{x(mz - ny) + y(nx - lz) + z(ly - mx)} \\ &= \frac{x dx + y dy + z dz}{\cancel{m x z} - \cancel{n x y} + \cancel{n x y} - \cancel{l y z} + \cancel{l y z} - \cancel{m x z}} \\ &= \frac{x dx + y dy + z dz}{0} \end{aligned}$$

$$\Rightarrow x dx + y dy + z dz = 0$$

Integrating

$$\Rightarrow \int x dx + \int y dy + \int z dz = \int 0$$

$$\Rightarrow \boxed{\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C} \text{ is second sol}$$

$\therefore$ The G.S. is $\phi(C_1, C_2) = 0$

$$\Rightarrow \phi\left(lx + my + nz, \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2}\right) = 0.$$

3) Solve $\frac{\partial^3 z}{\partial x^3} - 7 \frac{\partial^3 z}{\partial x \partial y^2} - 6 \frac{\partial^3 z}{\partial y^3} = e^{2x+y}$

Sol.: Given PDE in operator form is

$$(D^3 - 7DD'^2 - 6D'^3)Z = e^{2x+y}$$

The G.S. is $Z = Z_c + Z_p$

i, To find Z_c : Let $f(D, D') = D^3 - 7DD'^2 - 6D'^3$

put $D = m$; $D' = 1$

$$\Rightarrow f(m, 1) = 0 \Rightarrow m^3 - 7m - 6 = 0$$

$$\Rightarrow m = -1, 3, -2$$

$$\therefore Z_c = f_1(y + m_1 x) + f_2(y + m_2 x) + f_3(y + m_3 x)$$

$$\Rightarrow Z_c = f_1(y - x) + f_2(y + 3x) + f_3(y - 2x).$$

ii, to find z_p :

$$\begin{aligned}
 z_p &= \frac{1}{D^3 - 7DD'^2 - 6D'^3} e^{2x+y} \\
 &= \frac{1}{2^3 - 7(2)(1) - 6(1)} e^{2x+y} \quad \text{put } D=a=2, D'=b=1 \\
 &= \frac{1}{8-14-6} e^{2x+y} \\
 z_p &= -\frac{1}{12} e^{2x+y}
 \end{aligned}$$

$\therefore$ The G.Sol is $z = z_c + z_p$

$$\Rightarrow z = f_1(y-x) + f_2(y+3x) + f_3(y-2x) - \frac{1}{12} e^{2x+y}$$

4. Solve $(D^3 - 4D^2D' + 4DD'^2)z = 2 \sin(3x+2y)$

Sol:

Given PDE is

$$(D^3 - 4D^2D' + 4DD'^2)z = 2 \sin(3x+2y)$$

The G.Sol is $z = z_c + z_p$

i, to find z_c : Let $f(D, D') = D^3 - 4D^2D' + 4DD'^2$

put $D=m; D'=1$ in $f(D, D')$

$$\therefore f(m, 1) = 0 \Rightarrow m^3 - 4m^2 + 4m = 0$$

$$\Rightarrow m(m^2 - 4m + 4) = 0$$

$$\Rightarrow m(m-2)^2 = 0$$

$$\Rightarrow m = 0, 2, 2$$

$$\therefore z_c = f_1(y+0 \cdot x) + f_2(y+2x) + x f_3(y+2x)$$

ii, to find z_p : $z_p = \frac{1}{D^3 - 4D^2D' + 4DD'^2} 2 \sin(3x+2y)$

$$\text{put } D^2 = -3^2 = -9; D'^2 = -2^2 = -4$$

$$DD' = -ab = -3 \times 2 = -6$$

$$\begin{aligned}
 &= \frac{1}{-9D - 4D(-6) + 4D(-4)} 2 \sin(3x+2y) \\
 &= \frac{1}{-9D + 24D - 16D} 2 \sin(3x+2y)
 \end{aligned}$$

$$= \frac{1}{-25D+24D} \cdot 2 \sin(3x+2y)$$

$$= -\frac{1}{D} 2 \sin(3x+2y)$$

$$= -2 \cdot \frac{\partial}{\partial x} [\sin(3x+2y)]$$

$$z_p = -2 \cos(3x+2y) \cdot 3 = -6 \cos(3x+2y).$$

$\therefore$ the G.Sol is $z = z_c + z_p$

$$\Rightarrow z = f_1(y) + f_2(y+2x) + x f_3(y+2x) - 6 \cos(3x+2y).$$

5) Find the angle between the Normals to the surface $x^2 = yz$ at the points $(1,1,1)$ & $(2,4,1)$.

Sol.

Given surface $\phi = x^2 - yz = 0$

The normal to the surface $\phi = 0$ is $\nabla \phi$

$$\Rightarrow \nabla \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z}$$

$$\Rightarrow \nabla \phi = i(2x) + j(-z) + k(-y)$$

$$\Rightarrow \nabla \phi = 2xi - zj - yk$$

$$\text{Let } \vec{n}_1 = \nabla \phi_{(1,1,1)} = 2i - j - k$$

$$\vec{n}_2 = \nabla \phi_{(2,4,1)} = 4i - j - 4k$$

Let θ be the angle between $\vec{n}_1$ and $\vec{n}_2$

$$\therefore \theta = \cos^{-1} \left[\frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \right]$$

$$= \cos^{-1} \left[\frac{(2i - j - k) \cdot (4i - j - 4k)}{\sqrt{4+1+1} \sqrt{16+1+16}} \right]$$

$$= \cos^{-1} \left[\frac{8+1+4}{\sqrt{6} \sqrt{33}} \right]$$

$$\therefore \theta = \cos^{-1} \left[\frac{13}{\sqrt{6} \sqrt{33}} \right]$$

- 6) Find the Directional derivative of $\phi = xyz$ along the direction of the normal vector to the surface $f = x^2z + y^2x + yz^2 - 3 = 0$ at $(1,1,1)$.

Sol.

Given $\phi = xyz$

$$\Rightarrow \nabla \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} = yzi + xzj + xyk$$

$$\Rightarrow \nabla \phi_{(1,1,1)} = i + j + k$$

Given $f = x^2z + y^2x + yz^2 - 3 = 0$

the normal to the surface $f = 0$ is ∇f

$$\begin{aligned} \Rightarrow \nabla f &= i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \\ &= i [2xz + y^2] + j [2xy + z^2] + k [x^2 + 2yz] \end{aligned}$$

$$\Rightarrow \nabla f_{(1,1,1)} = 3i + 3j + 3k = \vec{a} \text{ (say)}$$

$\therefore$ the D.D of ϕ in the direction of $\vec{a} = \nabla \phi \cdot \frac{\vec{a}}{|\vec{a}|}$

$$= (i + j + k) \cdot \frac{(3i + 3j + 3k)}{|3i + 3j + 3k|}$$

$$= \frac{3+3+3}{\sqrt{9+9+9}} = \frac{9}{3\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

7. S.T the vector $\vec{F} = (x^2 - yz)i + (y^2 - zx)j + (z^2 - xy)k$ is irrotational and find its scalar potential.

Sol.

Given

$$\vec{F} = \underbrace{(x^2 - yz)}_{f_1} i + \underbrace{(y^2 - zx)}_{f_2} j + \underbrace{(z^2 - xy)}_{f_3} k$$

To P.T $\vec{F}$ is Irrotational:

$$\begin{aligned} \text{Curl } \vec{F} &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - yz & y^2 - zx & z^2 - xy \end{vmatrix} \\ &= i(-x + x) - j(-y + y) + k(-z + z) \\ &= \vec{0} \end{aligned}$$

$\therefore \text{curl } \vec{F} = \vec{0} \Rightarrow \vec{F}$ is Irrotational.

If $\vec{F}$ is Irrotational then $\exists$ a scalar potential function ϕ such that $\vec{F} = \nabla \phi$

$$\Rightarrow (x^2 - yz)\mathbf{i} + (y^2 - zx)\mathbf{j} + (z^2 - xy)\mathbf{k} = \mathbf{i} \frac{\partial \phi}{\partial x} + \mathbf{j} \frac{\partial \phi}{\partial y} + \mathbf{k} \frac{\partial \phi}{\partial z}$$

Equating coefficients on both sides we get

$$\frac{\partial \phi}{\partial x} = x^2 - yz; \quad \frac{\partial \phi}{\partial y} = y^2 - zx; \quad \frac{\partial \phi}{\partial z} = z^2 - xy$$

By total derivative of ϕ

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$= (x^2 - yz)dx + (y^2 - zx)dy + (z^2 - xy)dz$$

$$= x^2 dx + y^2 dy + z^2 dz - (yz dx + zx dy + xy dz)$$

$$d\phi = x^2 dx + y^2 dy + z^2 dz - d(xyz)$$

Integrating

$$\Rightarrow \int d\phi = \int x^2 dx + \int y^2 dy + \int z^2 dz - \int d(xyz)$$

$$\Rightarrow \boxed{\phi = \frac{x^3}{3} + \frac{y^3}{3} + \frac{z^3}{3} - xyz + C} \text{ is the reqd. scalar potential fn.}$$

8. S.T. $\text{Grad} \left[\text{Div} \left(\frac{\vec{r}}{r} \right) \right] = -\frac{2\vec{r}}{r}$ where $\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $|\vec{r}| = r$.

Sol. Given $\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $|\vec{r}| = r$

$$\Rightarrow |\vec{r}| = |x\mathbf{i} + y\mathbf{j} + z\mathbf{k}|$$

$$\Rightarrow r = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow r^2 = x^2 + y^2 + z^2 \longrightarrow \textcircled{1}$$

Diff $\textcircled{1}$ w.r.t x, y and z partially we get

$$\frac{\partial r^2}{\partial x} = \frac{2x}{r}; \quad \frac{\partial r^2}{\partial y} = \frac{2y}{r}; \quad \frac{\partial r^2}{\partial z} = \frac{2z}{r}$$

$$\text{Now } \frac{\bar{r}}{r} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r} = \frac{x}{r}\hat{i} + \frac{y}{r}\hat{j} + \frac{z}{r}\hat{k}$$

$$\Rightarrow \frac{\bar{r}}{r} = \frac{x}{r}\hat{i} + \frac{y}{r}\hat{j} + \frac{z}{r}\hat{k}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 $\frac{1}{r}$ $\frac{1}{r}$ $\frac{1}{r}$ $\frac{1}{r}$

$$\text{WKT } \text{Div } \vec{f} = \sum \frac{\partial f_i}{\partial x_i}$$

$$\Rightarrow \text{Div} \left(\frac{\bar{r}}{r} \right) = \sum \frac{\partial}{\partial x} \left[\frac{x}{r} \right]$$

$$= \sum \frac{r(1) - x \cdot \frac{\partial r}{\partial x}}{r^2}$$

$$= \sum \left[\frac{r}{r^2} - \frac{x}{r^2} \cdot \frac{x}{r} \right]$$

$$= \sum \left(\frac{1}{r} - \frac{x^2}{r^3} \right)$$

on Expanding

$$= \frac{3}{r} - \frac{1}{r^3} (x^2 + y^2 + z^2)$$

$$= \frac{3}{r} - \frac{1}{r^3} \cdot r^2$$

$$= \frac{3}{r} - \frac{1}{r} = \frac{2}{r}$$

$$\therefore \text{Div} \left(\frac{\bar{r}}{r} \right) = \frac{2}{r}$$

$$\text{WKT } \text{Grad } \phi = \sum \hat{i} \frac{\partial \phi}{\partial x_i}$$

$$\Rightarrow \text{Grad} \left(\text{Div} \frac{\bar{r}}{r} \right) = \text{Grad} \left(\frac{2}{r} \right) = \sum \hat{i} \frac{\partial}{\partial x_i} \left(\frac{2}{r} \right)$$

$$= \sum \hat{i} \cdot \left(-\frac{2}{r^2} \right) \frac{\partial r}{\partial x_i}$$

$$= \sum -\frac{2}{r^2} \hat{i} \cdot \frac{x}{r}$$

$$= -\frac{2}{r^3} \sum x \hat{i} = -\frac{2\bar{r}}{r^3}$$

$$\therefore \text{Grad} \left(\text{Div} \frac{\bar{r}}{r} \right) = -\frac{2\bar{r}}{r^3}$$

9) If $\vec{f} = x^2\vec{i} + y^2\vec{j}$ then Evaluate $\int_C \vec{f} \cdot d\vec{r}$ where C is the arc of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$

Sol.

Given $\vec{f} = x^2\vec{i} + y^2\vec{j}$

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$

$\vec{f} \cdot d\vec{r} = x^2 dx + y^2 dy$

$\therefore \int_C \vec{f} \cdot d\vec{r} = \int_C (x^2 dx + y^2 dy)$

$y = x^2 \Rightarrow \frac{dy}{dx} = 2x$

$\Rightarrow dy = 2x dx$

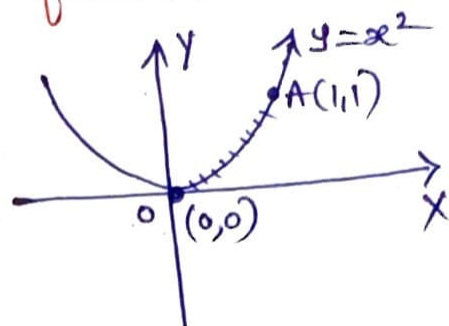
x its are from 0 to 1

$= \int_{x=0}^1 x^2 dx + x^4 \cdot 2x dx$

$= \int_{x=0}^1 (x^2 + 2x^5) dx$

$= \left(\frac{x^3}{3} + 2 \frac{x^6}{6} \right)_0^1$

$= \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$



10) Find the work done in moving the particle in the force field $\vec{f} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along the straight line from $(0,0,0)$ to $(1,2,3)$.

Sol.

Given $\vec{f} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$

Eq. of line Joining $P(0,0,0)$ to $Q(1,2,3)$ is

$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$

$$\Rightarrow \frac{x-0}{1-0} = \frac{y-0}{2-0} = \frac{z-0}{3-0}$$

$$\Rightarrow \frac{x}{1} = \frac{y}{2} = \frac{z}{3} = t \text{ (say)}$$

$$\Rightarrow x = t \quad y = 2t \quad z = 3t$$

$$\Rightarrow dx = dt \quad dy = 2dt \quad dz = 3dt$$

$$\text{At } P(0,0,0) : t = 0$$

$$Q(1,2,3) : t = 1$$

$\therefore$ The lts of t are $t = 0$ to 1

$$\therefore \text{Work done } W = \int_P^Q \vec{F} \cdot d\vec{a}$$

$$= \int_P^Q 3x^2 dx + (2xz - y) dy + z dz$$

$$= \int_{t=0}^1 3t^2 \cdot dt + (2t \cdot 3t - 2t) 2dt + 3t \cdot 3dt$$

$$= \int_{t=0}^1 (3t^2 + 12t^2 - 4t + 9t) dt$$

$$= \int_{t=0}^1 (15t^2 + 5t) dt$$

$$= \left[15 \frac{t^3}{3} + 5 \frac{t^2}{2} \right]_0^1$$

$$= \frac{15}{3} + \frac{5}{2}$$

$$= 5 + \frac{5}{2}$$

$$= \frac{15}{2}$$

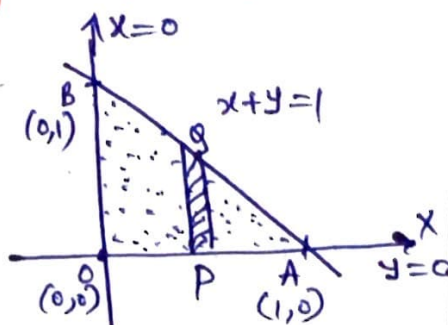
11. Verify Green's theorem for $\int (3x^2 - 8y^2) dx + (4y - 6xy) dy$ where C is the region bounded by $x=0$; $y=0$; $x+y=1$.

Sol: The curve C is bounded by

$$x=0; y=0; x+y=1$$

By Green's theorem

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$



$$\text{L.H.S} = \oint_C M dx + N dy = \int_{OA} + \int_{AB} + \int_{BO} \rightarrow \textcircled{1}$$

i, Along OA: Eq of OA is $y=0 \Rightarrow dy=0$
 x its are 0 to 1

$$\begin{aligned} \therefore \int_{OA} M dx + N dy &= \int_{x=0}^1 (3x^2 - 8y^2) dx + (4y - 6xy) dy \\ &= \int_{x=0}^1 3x^2 dx = 3 \left(\frac{x^3}{3} \right)_0^1 = 1 - 0 = 1 \end{aligned}$$

~~$\int_{BO} M dx + N dy =$~~

~~AB~~

ii, Along AB: Eq of AB is $x+y=1$
 $\Rightarrow y=1-x \Rightarrow dy=-dx$
 x its are $x=1$ to 0

$$\begin{aligned} \therefore \int_{AB} M dx + N dy &= \int_{x=1}^0 (3x^2 - 8y^2) dx + (4y - 6xy) dy \\ &= \int_{x=1}^0 [3x^2 - 8(1-x)^2] dx + [4(1-x) - 6x(1-x)](-dx) \\ &= \int_{x=1}^0 [3x^2 - 8(1-x)^2 - 4(1-x) + (6x - 6x^2)] dx \end{aligned}$$

$$\begin{aligned}
 &= \left[3 \frac{x^3}{3} - 8 \frac{(1-x)^3}{-3} - 4 \left(x - \frac{x^2}{2} \right) + \left(6 \frac{x^2}{2} - 6 \frac{x^3}{3} \right) \right]_1^0 \\
 &= \left[x^3 + \frac{8}{3} (1-x)^3 - 4 \left(x - \frac{x^2}{2} \right) + (3x^2 - 2x^3) \right]_1^0 \\
 &= \frac{8}{3} - \left\{ 1 + 0 - 4 \left(\frac{1}{2} \right) + (1) \right\} \\
 &= \frac{8}{3} - \{ 1 - 2 + 1 \} = \frac{8}{3}
 \end{aligned}$$

ii), Along B_0 : Eq of B_0 is $x=0 \Rightarrow dx=0$

the lts of y are $y=1$ to 0

$$\begin{aligned}
 \therefore \int_{B_0} M dx + N dy &= \int_{y=1}^0 (3x^2 - 8y^2) dx + (4y - 6xy) dy \\
 &= \int_{y=1}^0 4y dy = 4 \left(\frac{y^2}{2} \right)_1^0 = 2(0-1) = -2
 \end{aligned}$$

$$\therefore \int_C M dx + N dy = 1 + \frac{8}{3} - 2 = \frac{8}{3} - 1 = \frac{5}{3}$$

R.H.S: $\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

From given problem

$$\begin{array}{l|l}
 M = 3x^2 - 8y^2 & N = 4y - 6xy \\
 \frac{\partial M}{\partial y} = -16y & \frac{\partial N}{\partial x} = -6y
 \end{array}$$

Let PQ be the vertical strip

P lies on x -axis $\Rightarrow y=0$ (L.L of y)

Q lies on $x+y=1 \Rightarrow y=1-x$ (U.L of y)

$\therefore$ the lts of y are $y=0$ to $1-x$

the lts of x are $x=0$ to 1

$$\therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \int_{x=0}^1 \int_{y=0}^{1-x} (-6y + 16y) dy dx$$

$$= \int_{x=0}^1 \left[\int_{y=0}^{1-x} 10y \, dy \right] dx$$

$$= \int_{x=0}^1 10 \left(\frac{y^2}{2} \right)_0^{1-x} dx$$

$$= 5 \int_{x=0}^1 (1-x)^2 dx$$

$$= 5 \left[\frac{(1-x)^3}{-3} \right]_0^1 = -\frac{5}{3} [0 - 1] = \frac{5}{3}$$

$$\therefore L.H.S = R.H.S$$

Hence Green's Th is verified.

- 12) State Green's theorem and verify Green's theorem for $\int_C (xy + y^2) dx + x^2 dy$ where C is bounded by $y=x$ and $y=x^2$

Sol. statement: Green's theorem:

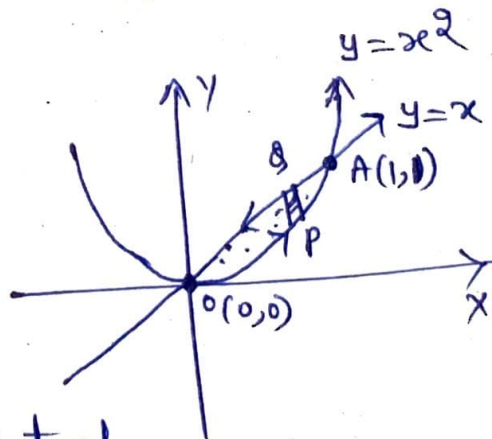
If R is a closed region in xy -plane bounded by a simple closed curve C and if M and N are continuous functions of x and y having continuous derivatives in R then $\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

Given $\int_C (xy + y^2) dx + x^2 dy$.

Given Curve C : $y=x \rightarrow \textcircled{1}$
 $y=x^2 \rightarrow \textcircled{2}$

Solving $\textcircled{1}$ & $\textcircled{2}$ we get

$O(0,0)$ and $A(1,1)$ are the points of intersections.



$$i, \int_C M dx + N dy = \int_{OA} + \int_{AO} \rightarrow \textcircled{1}$$

Along OA: Eq of OA is $y = x^2$
 $\Rightarrow dy = 2x dx$

the lts of x are $x = 0$ to 1

$$\begin{aligned} \therefore \int_{OA} (xy + y^2) dx + x^2 dy &= \int_{x=0}^1 (x \cdot x^2 + x^4) dx + x^2 \cdot 2x dx \\ &= \int_{x=0}^1 (x^3 + x^4 + 2x^3) dx \\ &= \int_{x=0}^1 (x^4 + 3x^3) dx \\ &= \left(\frac{x^5}{5} + 3 \frac{x^4}{4} \right)_0^1 = \frac{1}{5} + \frac{3}{4} = \frac{19}{20} \end{aligned}$$

Along AO: Eq of AO is $y = x \Rightarrow dy = dx$
 the lts of x are $x = 1$ to 0

$$\begin{aligned} \therefore \int_{AO} (xy + y^2) dx + x^2 dy &= \int_{x=1}^0 (x^2 + x^2) dx + x^2 dx \\ &= \int_{x=1}^0 3x^2 dx = 3 \left(\frac{x^3}{3} \right)_1^0 = (0 - 1) = -1 \end{aligned}$$

$$\therefore \text{L.H.S.} = \int_C M dx + N dy = \frac{19}{20} - 1 = -\frac{1}{20}$$

$$\text{R.H.S.} : \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

from given problem

$$\begin{array}{l|l} M = xy + y^2 & N = x^2 \\ \frac{\partial M}{\partial y} = x + 2y & \frac{\partial N}{\partial x} = 2x \end{array}$$

Set PQ be the vertical strip

P lies on $y = x^2$ (L.L of y)

Q lies on $y = x$ (U.L of y)

$\therefore$ the lts of y are $y = x^2$ to x

The lts of x are $x=0$ to 1

$$\therefore R.H.S = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \int_{x=0}^1 \int_{y=x^2}^x (2x - x - 2y) dx dy$$

$$= \int_{x=0}^1 \left[\int_{y=x^2}^x (x - 2y) dy \right] dx$$

$$= \int_{x=0}^1 \left(xy - \frac{2 \cdot y^2}{2} \right)_{x^2}^x dx$$

$$= \int_{x=0}^1 \left[(x \cdot x - x^2) - (x^3 - x^4) \right] dx$$

$$= \int_{x=0}^1 (x^2 - x^2 - x^3 + x^4) dx$$

$$= \left(\frac{x^5}{5} - \frac{x^4}{4} \right)_0^1$$

$$= \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}$$

$$\therefore L.H.S = R.H.S$$

Hence Green's theorem is verified.

Short Answer Questions:

1. Solve $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$

Sol. Given PDE is $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$

$$P = \sqrt{x}; \quad Q = \sqrt{y}; \quad R = \sqrt{z}$$

Auxiliary eqs are $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$$

①
②
③

$$\textcircled{1} = \textcircled{2} \Rightarrow \frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}}$$

Integrating both sides

$$\Rightarrow \int \frac{1}{\sqrt{x}} dx = \int \frac{1}{\sqrt{y}} dy$$

$$\Rightarrow 2\sqrt{x} = 2\sqrt{y} + C_1 \Rightarrow 2\sqrt{x} - 2\sqrt{y} = C_1$$

$$\Rightarrow \sqrt{x} - \sqrt{y} = C_1$$

$$\textcircled{2} = \textcircled{3} \Rightarrow \frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$$

Integrating both sides

$$\Rightarrow \int \frac{1}{\sqrt{y}} dy = \int \frac{1}{\sqrt{z}} dz$$

$$\Rightarrow 2\sqrt{y} = 2\sqrt{z} + C_2 \Rightarrow 2\sqrt{y} - 2\sqrt{z} = C_2$$

$$\Rightarrow \sqrt{y} - \sqrt{z} = C_2$$

$\therefore$ The G.Sol is $\phi(C_1, C_2) = 0$

$$\Rightarrow \phi(\sqrt{x} - \sqrt{y}, \sqrt{y} - \sqrt{z}) = 0$$

2) Solve $\frac{y^2 z}{x} p + xzq = y^2$

Sol. Given PDE is $\frac{y^2 z}{x} p + xzq = y^2$

$$P = \frac{y^2 z}{x}; \quad Q = xz; \quad R = y^2$$

The Auxiliary eqs are $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2}$$

$$\Rightarrow \frac{x dx}{y^2 z} = \frac{dy}{xz} = \frac{dz}{y^2} \quad \text{③}$$

①

②

$$\text{①} = \text{②} \Rightarrow \frac{x dx}{y^2 z} = \frac{dy}{xz}$$

$$\Rightarrow x^2 dx = y^2 dy$$

Integrate

$$\Rightarrow \int x^2 dx - \int y^2 dy = C_1$$

$$\Rightarrow \frac{x^3}{3} - \frac{y^3}{3} = C_1$$

$$\text{①} = \text{③} \Rightarrow \frac{x dx}{y^2 z} = \frac{dz}{y^2}$$

$$\Rightarrow x dx = z dz$$

Integrate

$$\Rightarrow \int x dx - \int z dz = C_2$$

$$\Rightarrow \frac{x^2}{2} - \frac{z^2}{2} = C_2$$

$\therefore$ The G.Sol is $\phi(C_1, C_2) = 0$

$$\Rightarrow \phi\left(\frac{x^3}{3} - \frac{y^3}{3}, \frac{x^2}{2} - \frac{z^2}{2}\right) = 0$$

3) solve $\frac{\partial^3 u}{\partial x^3} + 2 \frac{\partial^3 u}{\partial x^2 \partial y} - \frac{\partial^3 u}{\partial x \partial y^2} - 2 \frac{\partial^3 u}{\partial y^3} = 0$.

Sol.

The operator form is

$$(D^3 + 2D^2 D' - D D'^2 - 2D'^3) u = 0$$

The G.Sol is $u = u_c$

to find u_c : put $D = m$; $D' = 1$ then $f(m, 1) = 0$

$$\Rightarrow m^3 + 2m^2 - m - 2 = 0$$

$$\Rightarrow (m-1)(m+1)(m+2) = 0$$

$$\Rightarrow m = 1, -1, -2$$

$\therefore$ The sol is $u = u_c = f_1(y+x) + f_2(y-x) + f_3(y-2x)$

- 4) Define Gradient of a scalar point function and find Grad ϕ where $\phi = \log(x^2 + y^2 + z^2)$ at $(1, 1, 1)$

Sol. Gradient def: Let $\phi(x, y, z)$ be any scalar point function then the Gradient of ϕ is denoted by ~~Gradient of ϕ~~ $\text{grad } \phi$ or $\nabla \phi$ and is defined as

$$\text{Grad } \phi = \nabla \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} = \sum i \frac{\partial \phi}{\partial x}$$

Given $\phi = \log(x^2 + y^2 + z^2)$

$$\Rightarrow \frac{\partial \phi}{\partial x} = \frac{2x}{x^2 + y^2 + z^2} \Rightarrow \frac{\partial \phi}{\partial x} (1, 1, 1) = \frac{2}{3}$$

$$\frac{\partial \phi}{\partial y} = \frac{2y}{x^2 + y^2 + z^2} \Rightarrow \frac{\partial \phi}{\partial y} (1, 1, 1) = \frac{2}{3}$$

$$\frac{\partial \phi}{\partial z} = \frac{2z}{x^2 + y^2 + z^2} \Rightarrow \frac{\partial \phi}{\partial z} (1, 1, 1) = \frac{2}{3}$$

$$\therefore \text{Grad } \phi = \frac{2}{3}i + \frac{2}{3}j + \frac{2}{3}k$$

- 5) Find the unit normal vector to the surface $xy^3z^2 = 4$ at the point $(-1, -1, 2)$.

Sol. Given surface $\phi = xy^3z^2 - 4 = 0$

The normal vector to the surface $\phi = 0$ is $\nabla \phi$

$$\Rightarrow \nabla \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z}$$

$$\Rightarrow \nabla \phi = i[y^3z^2] + j[3xy^2z^2] + k[2xy^3z]$$

$$(-1, -1, 2) = i(-4) + j[-3(4)] + k[2(-1)(-1) \cdot 2]$$

$$= -4i - 12j + 4k$$

$$\therefore \text{unit normal vector} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{-4i - 12j + 4k}{\sqrt{16 + 144 + 16}} = \frac{-4i - 12j + 4k}{\sqrt{176}}$$

6) If $\vec{A} = 3xyz^2\vec{i} + 2xy^3\vec{j} - x^2yz\vec{k}$ find $(\vec{A} \times \nabla)\phi$ at $(1, -1, 1)$ and $\phi = 3x^2 - yz$

Sol. Given $\vec{A} = 3xyz^2\vec{i} + 2xy^3\vec{j} - x^2yz\vec{k}$

$$\phi = 3x^2 - yz$$

$$\Rightarrow \nabla\phi = i(6x) + j(-z) + k(-y)$$

$$\Rightarrow \nabla\phi = 6x\vec{i} - z\vec{j} - y\vec{k}$$

$$\text{Now } (\vec{A} \times \nabla)\phi = \vec{A} \times \nabla\phi$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3xyz^2 & 2xy^3 & -x^2yz \\ 6x & -z & -y \end{vmatrix}$$

$$= i[-2xy^4 - x^2yz^2] - j[-3xy^2z^2 + 6x^3yz] + k[-3xyz^3 - 12x^2y^3]$$

$$\Rightarrow (\vec{A} \times \nabla)\phi = i[-2+1] - j[-3-6] + k[3+12]$$

$$(1, -1, 1)$$

$$= -\vec{i} + 9\vec{j} + 15\vec{k}$$

7) Is the vector $\vec{f} = (x^2 + xy^2)\vec{i} + (y^2 + x^2y)\vec{j}$ Irrotational?

Sol. Given $\vec{f} = (x^2 + xy^2)\vec{i} + (y^2 + x^2y)\vec{j}$

$$\text{Curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + xy^2 & y^2 + x^2y & 0 \end{vmatrix}$$

$$= i[0 - 0] - j\left[\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(x^2 + xy^2)\right] + k\left[\frac{\partial}{\partial x}(y^2 + x^2y) - \frac{\partial}{\partial y}(x^2 + xy^2)\right]$$

$$= 0\vec{i} - 0\vec{j} + k[2xy - 2xy]$$

$$= \vec{0}$$

$\therefore \vec{f}$ is Irrotational

8) Define Divergence and Curl of a vector and find $\text{Div } \vec{F}$ and $\text{Curl } \vec{F}$ where $\vec{F} = \text{grad } (x^3 + y^3 + z^3 - 3xyz)$.

Sol.

Def: Divergence: If $\vec{F}$ is any continuously differentiable vector point function then Divergence of $\vec{F}$ is denoted by $\text{Div } \vec{F}$ (or) $\nabla \cdot \vec{F}$ and is defined as

$$\text{Div } \vec{F} \text{ (or) } \nabla \cdot \vec{F} = i \cdot \frac{\partial f}{\partial x} + j \cdot \frac{\partial f}{\partial y} + k \cdot \frac{\partial f}{\partial z}$$

If $\vec{F} = f_1 i + f_2 j + f_3 k$ then $\text{Div } \vec{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$.

Def: Curl of a vector: If $\vec{F}$ is any continuously differentiable vector point function then Curl $\vec{F}$ is denoted by $\text{Curl } \vec{F}$ (or) $\nabla \times \vec{F}$ and is defined as

$$\text{Curl } \vec{F} \text{ (or) } \nabla \times \vec{F} = i \times \frac{\partial f}{\partial x} + j \times \frac{\partial f}{\partial y} + k \times \frac{\partial f}{\partial z}$$

If $\vec{F} = f_1 i + f_2 j + f_3 k$ then

$$\text{Curl } \vec{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

Given $\vec{F} = \text{grad } (x^3 + y^3 + z^3 - 3xyz)$.

$$= i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z}$$

$$\Rightarrow \vec{F} = i \left[\underbrace{3x^2 - 3yz}_{f_1} \right] + j \left[\underbrace{3y^2 - 3xz}_{f_2} \right] + k \left[\underbrace{3z^2 - 3xy}_{f_3} \right]$$

$$\therefore, \text{Div } \vec{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= \frac{\partial}{\partial x} (3x^2 - 3yz) + \frac{\partial}{\partial y} (3y^2 - 3xz) + \frac{\partial}{\partial z} (3z^2 - 3xy)$$

$$= 6x + 6y + 6z$$

$$\text{Curl } \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 - 3yz & 3y^2 - 3xz & 3z^2 - 3xy \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (3z^2 - 3xy) - \frac{\partial}{\partial z} (3y^2 - 3xz) \right] - \hat{j} \left[\frac{\partial}{\partial x} (3z^2 - 3xy) - \frac{\partial}{\partial z} (3x^2 - 3yz) \right] + \hat{k} \left[\frac{\partial}{\partial x} (3y^2 - 3xz) - \frac{\partial}{\partial y} (3x^2 - 3yz) \right]$$

$$= \hat{i} [-3x + 3x] - \hat{j} [-3y + 3y] + \hat{k} [-3z + 3z]$$

$$= 0 \cdot \hat{i} + 0 \cdot \hat{j} + 0 \cdot \hat{k} = \vec{0}$$

9) If $\vec{f} = (x+3y)\hat{i} + (y-2z)\hat{j} - (x+pz)\hat{k}$ is solenoidal find P.

Sol. Given $\vec{f} = \underbrace{(x+3y)}_{f_1}\hat{i} + \underbrace{(y-2z)}_{f_2}\hat{j} - \underbrace{(x+pz)}_{f_3}\hat{k}$

$\therefore \vec{f}$ is solenoidal $\Rightarrow \text{Div } \vec{f} = 0$

$$\Rightarrow \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} = 0$$

$$\Rightarrow 1 + 1 - p = 0 \Rightarrow \boxed{p = 2}$$

10) If $\vec{a}$ is a constant vector P.T $\text{Grad}(\vec{a} \cdot \vec{r}) = \vec{a}$ where

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Sol. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\Rightarrow \vec{a} \cdot \vec{r} = a_1x + a_2y + a_3z$$

~~$\Rightarrow \text{grad}(\vec{a} \cdot \vec{r}) =$~~

wk $\text{grad } \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$

~~$\Rightarrow \text{grad}(\vec{a} \cdot \vec{r}) = \hat{i} \frac{\partial}{\partial x} (a_1x + a_2y + a_3z) + \hat{j} \frac{\partial}{\partial y} (a_1x + a_2y + a_3z) + \hat{k} \frac{\partial}{\partial z} (a_1x + a_2y + a_3z)$~~

$$\Rightarrow \text{grad}(\vec{a} \cdot \vec{r}) = \hat{i} \cdot a_1 + \hat{j} \cdot a_2 + \hat{k} \cdot a_3$$

$$= a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\therefore \text{Grad}(\vec{a} \cdot \vec{r}) = \vec{a}$$

11. In what direction from the point $(1, -1, 3)$ the directional derivative of $\phi = 2xy + z^2$ is maximum and what is the magnitude of this Maximum?

Sol.

Given $\phi = 2xy + z^2$

$$\Rightarrow \nabla \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z}$$
$$= i(2y) + j(2x) + k(2z).$$

$$\Rightarrow \nabla \phi_{(1, -1, 3)} = -2i + 2j + 6k$$

The D.D of ϕ is Max in the direction of The Normal vector to the surface $\phi = 0$ at a point P. and The Mag. of this Max is $|\nabla \phi| = |-2i + 2j + 6k| = \sqrt{4 + 4 + 36} = \sqrt{44} = 2\sqrt{11}.$

12. Evaluate $\int y^2 dx - 2x^2 dy$ along the parabola $y = x^2$ from $(0, 0)$ to $(2, 4)$.

Sol.

Given parabola $y = x^2$

$$\Rightarrow \frac{dy}{dx} = 2x$$

$$\Rightarrow dy = 2x dx$$

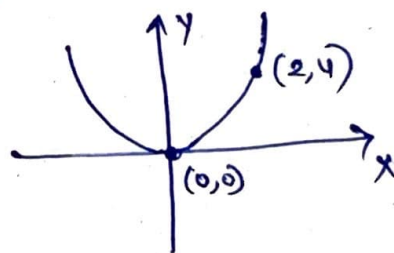
The lts of x are $x = 0$ to 2

$$\therefore \int y^2 dx - 2x^2 dy = \int_{x=0}^2 (x^2)^2 dx - 2x^2 \cdot 2x dx$$

$$= \int_{x=0}^2 (x^4 - 4x^3) dx$$

$$= \left(\frac{x^5}{5} - 4 \frac{x^4}{4} \right) \Big|_0^2$$

$$= \frac{2^5}{5} - 2^4$$



13. State Green's theorem:

If R is a closed region in xy -plane bounded by a simple closed Curve 'C' and if M and N are Continuous functions of x and y having Continuous partial derivatives in R then

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy.$$

14. State Stokes's theorem:

Let S be an open surface bounded by a closed, non intersecting Curve 'C'. If $\vec{F}$ is any differentiable vector point function then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} ds$$

where $\hat{n}$ is unit outward normal vector at any point of the surface.

15. State Gauss Divergence theorem:

Let S be a closed surface enclosing a volume V .

If $\vec{F}$ is a continuously differentiable vector point function

then $\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \text{Div } \vec{F} dv$ where $\hat{n}$ is the outward

normal vector drawn at any point of S .